HB - Rachel (hb_integration observer) — implemented equations (M1–M5)

Everything lives on the discrete direction grid $\theta \in$ **$\{1, \ldots, 360\}^\circ$**, with the prior mode μ **$= 225^\circ$**. The building block is the (normalised) von Mises pmf:

```
V(θ; μ, κ) = exp(κ · cos(θ − μ)) / Σ_θ' exp(κ · cos(θ' − μ))
```

κ (kappa) is the concentration — large κ = narrow bump, small κ = broad.

M1 — Mixed hyper-prior (the model's defining object)

A peaked von Mises at the prior mean, unioned with a uniform floor:

```
p(θ | κ, α) = α · V(θ; 225, κ)  +  (1 − α) · (1/360)
```

- α = mixture weight ("does the prior structure apply at all")
- κ = prior concentration (how tight the belief about 225° is)

Source: `mixture_prior()`.

M2 — Sensory likelihood + measurement model (per coherence c)

With sensory concentration k_like(c) (one per coherence 0.06 / 0.12 / 0.24):

```
measurement:   P(m | θ_true) = V(m; θ_true, k_like(c))
likelihood:    P(θ | m)      = V(θ; m,      k_like(c))
```

M3 — Posterior, MAP read-out, motor noise (per fixed κ)

For each internal measurement m, form the posterior, take its mode (ties shared equally), then marginalise over the measurement to get the read-out for that κ:

```
posterior(θ | m)  ∝  P(θ | m) · p(θ | κ, α)

R_κ(est | θ_true) = Σ_m  P( est = mode[ posterior(· | m) ] ) · P(m | θ_true)
```

Each per-κ read-out is then circularly convolved with von Mises motor noise V(· ; 0, k_motor). Source: `_map_readout()` + motor convolution.

M4 — Estimate distribution = belief-average of read-outs, then lapse

THE key structural choice (read-out-then-average): average the per-κ read-outs over the current belief b_t(κ), then mix in the uniform lapse:

```
                [ Σ_κ b_t(κ) · R_κ(est | θ_true) ]  +  p_random · (1/360)
p_t(est | ·) =  -------------------------------------------------------
                                  1 + p_random
```

Source: `estimate_distribution()` (percept = belief @ readouts).

M5 — Trial-by-trial belief update over κ (the "learning of prior confidence")

Between trials the belief leaks toward its initial state (volatility), then is corrected by the feedback direction θ_fb shown at trial end:

```
predict (forget):   b~_t(κ) = (1 − λ) · b_{t−1}(κ)  +  λ · b_0(κ)

correct (Bayes):    b_t(κ)  ∝  b~_t(κ) · [ α · V(θ_fb; 225, κ) + (1 − α)/360 ]
                   then renormalise so Σ_κ b_t(κ) = 1
```

- λ (lambda) = volatility / forgetting. $\lambda = 0 \to$ pure accumulation; $\lambda = 1 \to$ belief resets every trial. Source: `update_belief()` = `forget()` then `bayes_correct()`.

Fitted parameters (7)

```
k_like(0.06), k_like(0.12), k_like(0.24)   sensory reliability, per coherence
α                                          prior mixture weight (fixed across blocks)
```

`k_motor`	`motor-noise concentration`
`p_random`	`lapse rate`
`λ`	`volatility / forgetting`

κ is NOT fitted — it is the *learned latent*, carried as the belief b_t(κ) over a ~15-point κ grid and updated every trial by M5. (N_PARAMS = 7.)

Two things the equations encode that are easy to miss

Bimodality is EMERGENT, not switched. In M3, when the measurement lands near 225° the von Mises term of M1 dominates (estimate pulled to the prior); when it lands far, the uniform floor dominates (estimate at the evidence). Sweeping m yields two modes with NO switch statement — the "switch" is a *derived responsibility* (which mixture component of M1 explains m).

Why M4 averages read-outs, not priors. It marginalises κ AFTER the read-out (Σ_κ b(κ)·R_κ), not by collapsing the belief into one prior first. This is deliberate: it matches the online switching observer's convention so that the switch-vs-integration comparison differs ONLY in the read-out rule. (This is exactly the axis on which the branch's independent model diverges — it integrates into ONE posterior before reading out.)